

HAO LIU

Applied Scientist — Agentic AI Systems — Post-Training for Tool-Using LLMs — Ph.D. in Computer Science

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PROFESSIONAL SUMMARY

Applied Scientist with a Ph.D. in Computer Science building production agentic systems and post-training for tool-using LLMs. Experience spans autonomous coding and DevOps agents, including orchestration, context gathering, evaluation-driven system improvement, and SFT/RL-based training for tool use, multi-step reasoning, and environment interaction. Published at ICLR, NeurIPS, and WWW, with a track record of translating research into deployed systems.

WORK EXPERIENCE

Applied Scientist

Seattle, WA

Amazon Web Services

July 2025 – Present

AWS DevOps Agent

- Building post-training pipelines for a DevOps investigation agent, focused on improving tool use, reasoning, and multi-step decision making in operational workflows.
- Built an SFT training pipeline and real-trajectory-based data curation workflow for agent behaviors including tool selection, schema-constrained invocation, result interpretation, and next-step intent generation.
- Developing RL-based training workflows for end-to-end agent interaction with tools and environment feedback in log investigation scenarios.

Kiro Autonomous Agent

- Built and shipped core orchestration workflows for a launched autonomous coding agent, spanning request understanding, repository grounding, plan generation, and delegated execution, in collaboration with applied scientists and engineers.
- Owned a delegation framework that translated plans into structured task execution, analyzed sub-agent results, and adaptively revised next actions in the execution loop.
- Investigated quality degradation in planning-based agent workflows, traced a major failure mode to information loss between planner and executor, and improved task success from 50% to 54%.
- Developed a context-gathering code exploration subagent that retrieved and summarized relevant repository context to help the main agent generate better plans when initial information was insufficient.

EDUCATION

Washington University in St. Louis

Ph.D. in Computer Science & Engineering

St. Louis, MO

Sep. 2019 – May 2025

Beijing Normal University

B.S. in Mathematics

Beijing, China

Sep. 2015 – June 2019

SELECTED RESEARCH

GOFA: A Generative One-For-All Model for Joint Graph Language Modeling

ICLR 2025

- Designed a graph foundation model that integrates GNN layers with LLM layers, leveraging the generative strengths of LLMs for free-form output and the structural learning capabilities of GNNs.
- Pioneered a novel pretraining paradigm focused on graph-level next-token prediction, facilitating large-scale self-supervised learning on various text-attributed graphs.
- Achieved state-of-the-art zero-shot performance on six node/link/graph-level datasets after instruction fine-tuning on a small number of data, demonstrating its potential as a foundation model.

One for All: Training One Graph Model for All Classification Tasks

ICLR 2024 Spotlight

- Developed the first graph foundation model capable of handling multiple classification tasks across various datasets and domains in supervised, few-shot, and zero-shot scenarios.

- Employed Large Language Models to standardize and unify graph data representation, transforming features of diverse graphs into consistent natural language formats, enabling multi-dataset training in the graph domain.
- Innovated a graph prompting paradigm, advancing the model's in-context learning capabilities.

Unsupervised Meta-Learning for Transductive Few-shot Node Tasks

WWW 2024

- Designed the first unsupervised meta-learning method for transductive few-shot node classification.
- Innovated an unsupervised meta-task construction method by leveraging the strengths of contrastive learning, enabling comprehensive use of graph nodes in the learning process.
- Achieved state-of-the-art performance on seven datasets, with at least 11.18% and up to 20.56% absolute accuracy improvement over existing meta-learning baselines.

PUBLICATIONS

Conference:

- **GOFA: A Generative One-For-All Model for Joint Graph Language Modeling**
L. Kong*, J. Feng*, **Hao Liu***, C. Huang, J. Huang, Y. Chen, M. Zhang
The Thirteenth International Conference on Learning Representations (**ICLR 2025**)
- **One for All: Towards Training One Graph Model for All Classification Tasks**
Hao Liu*, J. Feng*, L. Kong*, N. Liang, D. Tao, Y. Chen, M. Zhang
The Twelfth International Conference on Learning Representations (**ICLR 2024 Spotlight**)
- **Graph Contrastive Learning Meets Graph Meta Learning: A Unified Method for Few-shot Node Tasks**
Hao Liu, J. Feng, L. Kong, D. Tao, Y. Chen, M. Zhang
The Web Conference 2024 (**WWW 2024**)
- **TabContrast: A Local-Global Level Method for Tabular Contrastive Learning**
Hao Liu, Y. Chen, B. Fritz, C. King
NeurIPS 2023 Second Table Representation Learning Workshop (**NeurIPS 2023 TRL**)
- **MAG-GNN: Reinforcement Learning Boosted Graph Neural Network**
L. Kong, J. Feng, **Hao Liu**, D. Tao, Y. Chen, M. Zhang
Thirty-seventh Conference on Neural Information Processing Systems (**NeurIPS 2023**)
- **Extending the Design Space of Graph Neural Networks by Rethinking Folklore Weisfeiler-Lehman**
J. Feng, L. Kong, **Hao Liu**, D. Tao, F. Li, M. Zhang, Y. Chen
Thirty-seventh Conference on Neural Information Processing Systems (**NeurIPS 2023**)

Preprint:

- **TAGLAS: An Atlas of Text-attributed Graph Datasets in the Era of Large Graph and Language Models**
J. Feng, **Hao Liu***, L. Kong*, Y. Chen, M. Zhang
- **Time Associated Meta Learning for Clinical Prediction**
Hao Liu, M. Zhang, Z. Dong, L. Kong, Y. Chen, B. Fritz, D. Tao, C. King

(* indicates equal contribution)

TECHNICAL SKILLS

Programming: Python, C, Matlab

ML / LLM Frameworks: PyTorch, PyTorch Lightning, vLLM, TRL, VeRL, Scikit-learn, PyG

Agentic AI & Post-Training: Agent orchestration, tool-use workflows, SFT, RL-based training workflows, evaluation

Data / Developer Tools: Pandas, NumPy, Docker, Git, Jupyter, VS Code, Google Cloud Platform